

SCHOOL OF SCIENCE AND ENGINEERING
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING



EGE322: Electronics I Lab

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Lab #8 - Current Mirror

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Table Of Contents

Introduction	3
Theory	3
Procedure	7
Week 1	7
Part I. Current Mirror Analysis and Simulation	7
Part II. Current Mirror Physical Verification	8
Part III. Emitter Degeneration Analysis and Simulation.....	8
Part IV. Emitter Degeneration Physical Verification.....	9
Week 2	9
Part I. Wilson Current Mirror Analysis and Simulation.....	9
Part II. Current Mirror Comparison.....	10
Part III. Wilson Current Mirror Physical Verification.....	11
Results.....	11
Week 1	11
Part I. Current Mirror Analysis and Simulation	11
Part II. Current Mirror Physical Verification	12
Part III. Emitter Degeneration Analysis and Simulation.....	13
Part IV. Emitter Degeneration Physical Verification.....	13
Week 2	14
Part I. Wilson Current Mirror Analysis and Simulation.....	14
Part II. Current Mirror Comparison.....	15
Part III. Wilson Current Mirror Physical Verification.....	16
Conclusion.....	17

Introduction

This experiment aimed to examine and understand three configurations of NPN BJT current mirrors. Each current mirror differed in effectiveness, as each subsequent current mirror proposed a design which corrected some of the imperfections in the previous one. This experiment was divided into two weeks. In the first week, a basic configuration of the current mirror and an updated configuration with emitter degeneration resistors were introduced and tested. In the second week, an improved current mirror, known as the Wilson current mirror, was introduced for experimentation. Each configuration was simulated in LTspice and constructed on a breadboard. The reference and the load currents, measured by the Agilent Tools, were observed and compared.

Theory

Current mirrors create an input current source, the input reflects or swaps the direction of flow this current to produce a current sync the current exiting the mirror/circuit. As a result, we obtain a current sync which pulls current from the circuit. The ideal current sink provides that

$$\beta I_B = I_{Ref}$$

That means

$$\beta I_B = I_L$$

However, in this case the V_{CE} and V_{EE} are fixed values so only the left side of the circuit experiences the ideal case where I_B equals out at a current value of I_{ref} after .3V which is the “active region”. V_L and I_L on the right side of the current are not fixed which results in the early effect. The early effect behavior is displayed in **Figure 1**.

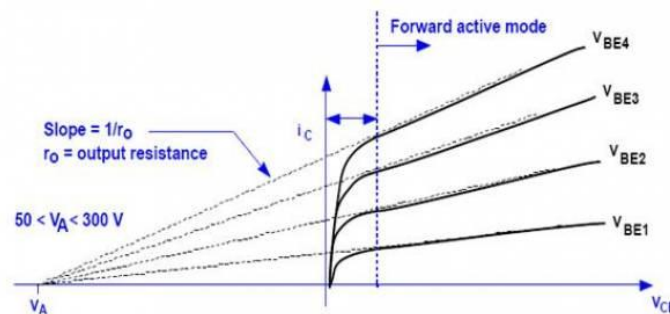


Figure 1

A current mirror is a circuit configuration that replicates a reference current. The basic configuration of a current mirror is shown in **Figure 2**. Assuming the ideal conditions when both BJTs in the circuit have the same β and V_{BE} , the relationship between I_L and I_{ref} when the BJT Q2 is in the active mode can be derived as follows.

$$I_{ref} = I_{C1} + I_1 = I_{C1} + I_{B1} + I_{B2} = \beta_1 I_{B1} + I_{B1} + I_{B2} = (\beta + 2)I_{B2} = \frac{\beta+2}{\beta} I_L$$

$$\Rightarrow I_L = \frac{\beta}{\beta+2} I_{ref} \quad \text{(Equation 1)}$$

When β is very high, we have $\beta \approx \beta + 2$. So, $I_L \approx I_{ref}$.

However, simulations and measurements of the physical implementations of this circuit do not demonstrate the theoretical relationship of I_{ref} and I_L . This is because of four non-ideal factors acting on Q2: the Early effect, V_{BE} variation, thermal effect, and β variation.

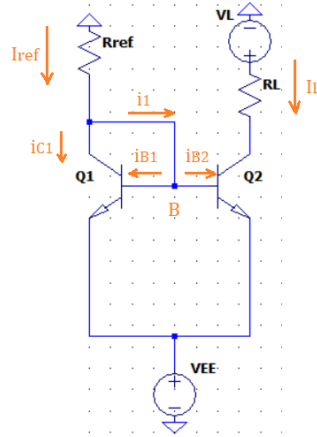


Figure 2

β Variation

In reality, the value of β for each BJT of the same model varies. This affects the accuracy of the current replication according to **Equation 1**.

The Early Effect

The Early effect is also known as the base-width modulation effect. When V_{BE} of a BJT is fixed, increasing V_{CE} increases the reverse-bias voltage on the collector-base junction, which increases the width of the depletion region of this junction. This causes a decrease in the effective base

width W , and thus an increase in I_S because I_S is inversely proportional to W . By **Equation 2**, an increase in I_S results in an increase in I_L .

$$I_L = I_C = I_S e^{V_{BE}/V_T} \quad (\text{Equation 2})$$

V_{BE} Variation and the Thermal Effect

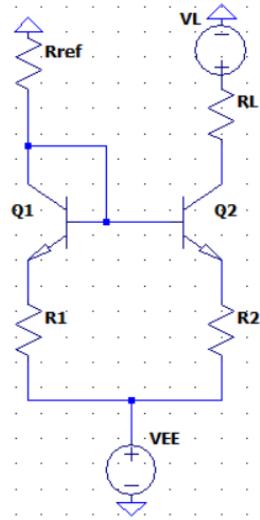
In a physical circuit, the V_{BE} of the two BJTs may vary. As shown in **Figure 2**, the emitters of both BJTs are connected to the same voltage source and the bases are connected to the common node B. Different V_{BE} values create different voltages at the node B. This induces a current conflict at the node B and results in heat dissipation on Q2. From **Equation 2**, we have

$$V_{BE} = V_T \ln(I_C/I_S)$$

Increasing temperature increases V_T and thus increases V_{BE} . This increases I_C , which increases V_{BE} further. This process is known as the thermal effect. So, the variation of V_{BE} and the thermal effect increase I_L . An increasing I_L will increase temperature, thus increasing V_T and cause the value of I_L to continue increasing over time, creating an ongoing cycle where I_L will never stabilize.

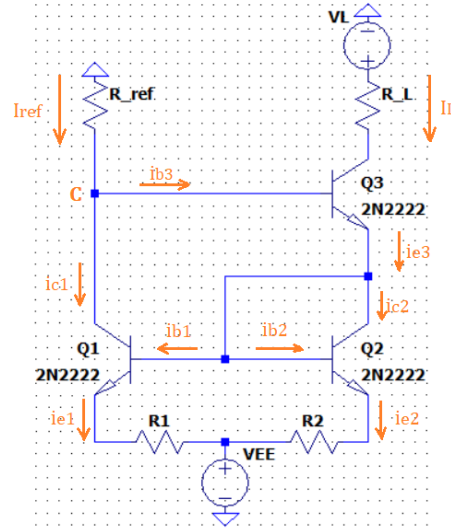
This experiment's objective was to minimize the above undesired effects and create a current mirror with a stable and accurate output I_L . The configuration in **Figure 3a** was derived by adding a resistor, called the emitter degeneration resistor (EDR), between each emitter and the voltage source V_{EE} in the circuit shown in **Figure 2**. Realistically, the V_{BE} of the two BJTs are different, the two resistors take on part of the voltage to eliminate the voltage difference at the common node B. This removes the effect of variation of V_{BE} and thus reduces the thermal effect significantly.

The configuration in **Figure 3b** is called a BJT Wilson Current Mirror. It includes a third BJT Q3 above a basic current mirror with EDR. This configuration ensures the voltage at the node C to be fixed at $V_{EE} + V_{BE2} + V_{BE3}$, which limits V_{CE1} to a fixed value. This removes the Early effect so that I_L stays constant as V_L changes.



a. Basic with Emitter

Degeneration Resistors



b. Wilson

Figure 3

Assuming $R_1 = R_2$ and ideal conditions for all the BJTs in the Wilson Current Mirror, the relationship between I_L and I_{ref} can be derived as follows.

$$V_{BE1} + i_{e1}R_1 = V_B - V_{EE} = V_{BE2} + i_{e2}R_2$$

$$\Rightarrow i_{e1} = \frac{R_1}{R_2}i_{e2} + \frac{V_{BE1} - V_{BE2}}{R_2}$$

$$\Rightarrow i_{e1} = i_{e2}$$

$$\Rightarrow i_{b1} = i_{b2} \text{ and } i_{c1} = i_{c2}$$

$$I_L = \frac{\beta}{1+\beta}i_{e3} = \frac{\beta}{1+\beta}(i_{c2} + i_{b2} + i_{b1}) = \frac{\beta}{1+\beta}(\beta i_{b2} + i_{b2} + i_{b1}) = \frac{\beta(2+\beta)}{1+\beta}i_{b1} = \frac{2+\beta}{1+\beta}i_{c1}$$

$$I_{ref} = i_{c1} + i_{b3} = i_{c1} + \frac{1}{\beta}I_L = \frac{1+\beta}{2+\beta}I_L + \frac{1}{\beta}I_L = \frac{\beta^2 + 2\beta + 2}{\beta^2 + 2\beta}I_L$$

$$I_L = \frac{\beta^2 + 2\beta}{\beta^2 + 2\beta + 2}I_{ref}$$

The term β^2 in the coefficient of I_{ref} reduces the effect of β variation significantly.

Procedure

This experiment utilized the following electrical components: 100 Ω , 150 Ω , 1k Ω , 1.5k Ω , and 12k Ω resistors, and 2N2222 NPN transistors. The Keysight E3630A Triple Output DC Power Supply was used as a DC voltage source for V_{CE} and V_{EE} . The measured voltages were visualized in Agilent Measurement Manager through the oscilloscope in Keysight U2781A.

Week 1

In this week, the main focus was to analyze, simulate and physically implement the basic configurations of the current mirror.

Part I. Current Mirror Analysis and Simulation

The experiment started by analyzing the provided circuit as shown in **Figure 1**. The theoretical relationship $I_L = f(I_{ref})$ and the range of V_L for Q2 to work in the active mode were calculated. The value of R_{ref} was determined provided that I_{ref} was 1mA, V_{ref} was 0V, V_{EE} was -15V and V_{BE} was 0.7V for both of the transistors. The resistor R_L was set to 1k Ω .

After the theoretical analysis and calculations, the circuit was simulated in LTspice. The voltage V_L was set to a DC Sweep ranging from ($V_{L(min)} - 5V$) to 20V with a step of 0.01V. The current I_{ref} and I_L were measured and compared to week 1. The simulated circuit is shown in **Figure 4** below.

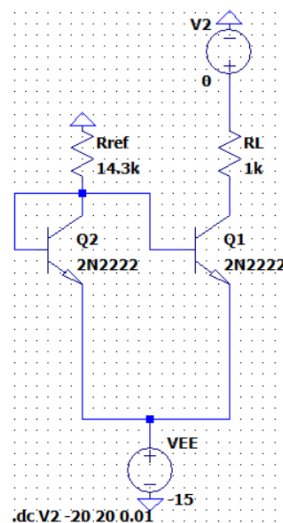


Figure 4

Part II. Current Mirror Physical Verification

The current mirror was constructed on a breadboard as shown in **Figure 5**.

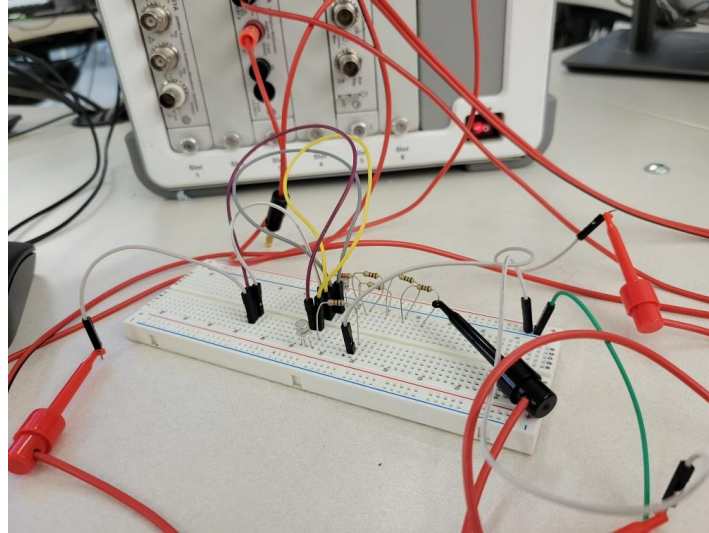


Figure 5

V_L was connected to DC 5V, and I_{ref} was measured first. Then the DC voltage source was turned off when the circuit was modified to measure I_L . I_L was measured after the DC voltage source was turned back on. Leaving all conditions static, I_L was observed over a period of time. The results were compared to the theoretical and simulated ones.

Part III. Emitter Degeneration Analysis and Simulation

Two resistors R_1 and R_2 were each placed between a BJT and V_{EE} as indicated in **Figure 6**.

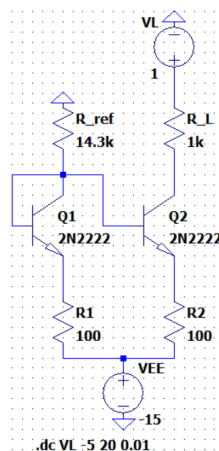


Figure 6

The circuit was analyzed and the relationship between the two emitter currents were derived. After setting R_1 and R_2 to 100Ω , the circuit was simulated under the same DC sweep as in Part I. The currents I_{ref} and I_L were measured, observed and compared.

Part IV. Emitter Degeneration Physical Verification

The circuit was constructed on a breadboard as shown in **Figure 7**.

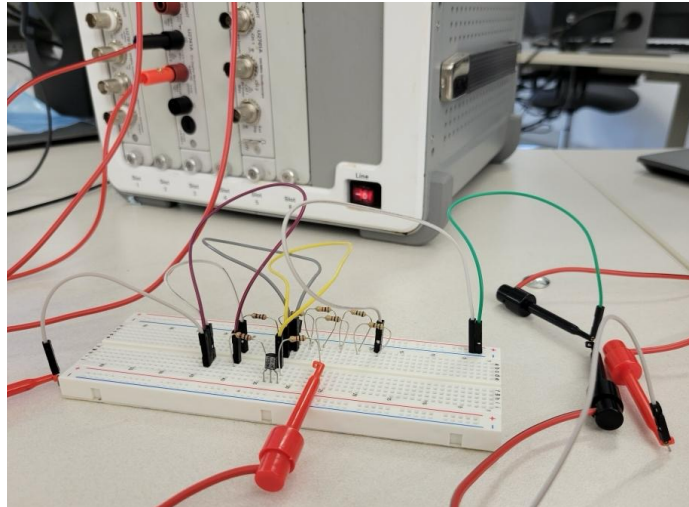


Figure 7

V_L was connected to DC 5V, and I_{ref} was measured first. Then the DC voltage source was turned off when the circuit was modified to measure I_L . I_L was measured after the DC voltage source was turned back on. Leaving all conditions static, I_L was observed over a period of time. Next, V_L was connected to a separate DC voltage source. I_L was measured when V_L was set to -20V, -17V, -13V, -5V, 0V, 5V, 13V, 17V and 20V. The results were compared to the simulated ones.

WEEK 2

In this week, the main focus was to analyze, simulate and physically implement the Wilson Current Mirror.

Part I. Wilson Current Mirror Analysis and Simulation

The experiment started by analyzing the provided circuit as shown in **Figure 3b**. The theoretical relationship $I_L = f(I_{ref})$ and the range of V_L for Q3 to work in the active mode were calculated. The results were compared to week 1. The value of R_{ref} was determined provided that I_{ref} was

1mA, V_{ref} was 0V, V_{EE} was -15V and V_{BE} was 0.7V for all the transistors. The resistors R_1 and R_2 were set to 100Ω , and R_L was set to $1k\Omega$.

After the theoretical analysis and calculations, the circuit was simulated in LTspice. The voltage V_L was set to a DC Sweep ranging from ($V_{L(min)} - 5V$) to 20V with a step of 0.01V. The current I_{ref} and I_L were measured and compared to week 1. The simulated circuit is shown in **Figure 8** below.

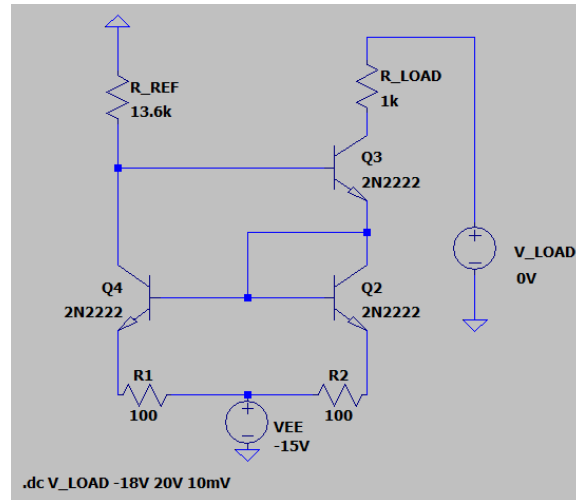


Figure 8

Part II. Current Mirror Comparison

All the three configurations were constructed in one LTspice file sharing common V_L and V_{EE} , as shown in **Figure 9**. Then they were simulated under the DC Sweep of V_L ranging from 0V to 20V with a step of 0.01V. The three I_L were measured, observed and compared.

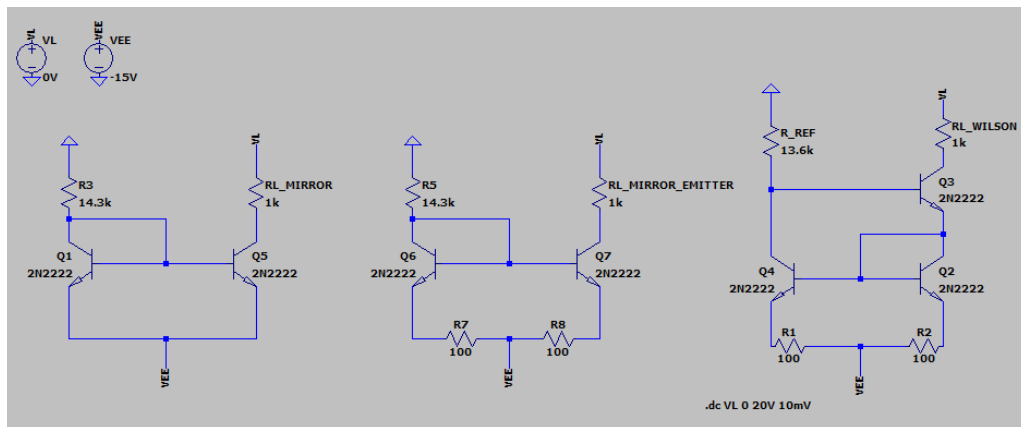


Figure 9

Part III. Current Mirror Physical Verification

The Wilson Current Mirror was constructed on a breadboard as shown in **Figure 10**.

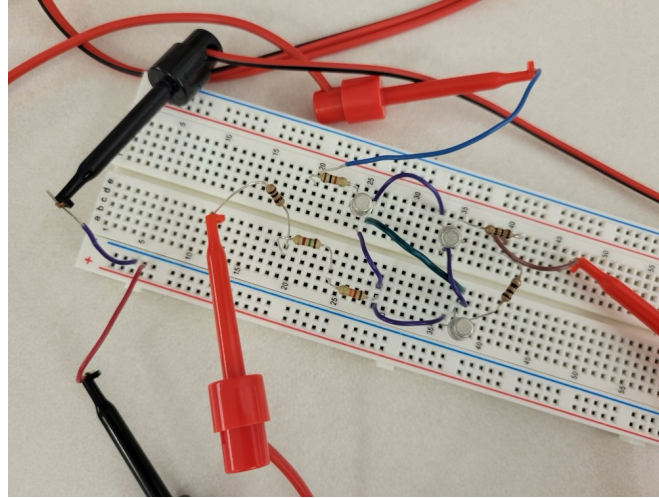


Figure 10

V_L was connected to DC 5V, and I_{ref} was measured first. Then the DC voltage source was turned off when the circuit was modified to measure I_L . I_L was measured after the DC voltage source was turned back on. Leaving all conditions static, I_L was observed over a period of time. Next, V_L was connected to a separate DC voltage source. I_L was measured when V_L was set to -20V, -17V, -13V, -5V, 0V, 5V, 13V, 17V and 20V. The results were compared to the simulated ones.

Results

Week 1

Part I. Current Mirror Analysis and Simulation

(a). As calculated in Theory, $I_L = \frac{\beta}{\beta+2} I_{ref}$

When β is very high, we have $I_L \approx I_{ref}$. Assume $\beta = 200$, then $I_L = 0.9901 I_{ref}$.

(b). The domain of the function V_L is calculated by ensuring that Q2 works in the active mode, which is $V_{CE} > 0.3V$. So, we have

$$V_C - V_E = V_L - R_L I_L - V_{EE} > 0.3$$

$$\Rightarrow V_L > 0.3 + R_L I_L + V_{EE}$$

$$\Rightarrow V_L > 0.3 + \frac{\beta}{\beta+2} R_L I_{ref} + V_{EE}$$

(c). $V_{C1} = V_B = V_{BE} + V_{EE} = 0.7 - 15 = -14.3[V]$

$$R_{ref} = (V_{ref} - V_{C1})/I_{ref} = (0 + 14.3)/1 = 14.3[k\Omega]$$

(d). The smallest V_L that the system can operate under is

$$V_{L(min)} = 0.3 + \frac{200}{202} * 1 * 1 - 15 = -13.71[V]$$

(e). As shown in **Figure 11**, I_{ref} was rather stable, ranging from 1.007mA to 1.003mA. I_L started at -3.6mA, increased linearly to 0.987mA since Q2 was in the saturation mode. As Q2 entered active mode, I_L mirrored I_{ref} with slow increase as explained in Theory.

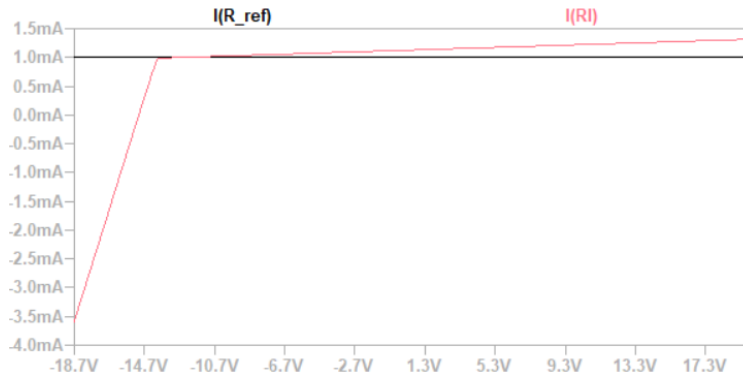


Figure 11

Part II. Current Mirror Physical Verification

In the physical circuit, I_{ref} was measured to be stable at 1.0003mA while I_L was 1.5765mA initially and then kept increasing to above 3mA over time. The result observed in the physical circuit was different from that of the simulated and theoretical ones. When V_L was fixed, both the theoretical and the simulated I_L were a constant. This is because the theoretical and the simulated calculations assumed an ideal condition, i.e. none of the four factors explained in the Theory section were considered. However, in real-life experiments those factors existed and lead to an increasing I_L .

Part III. Emitter Degeneration Analysis and Simulation

(b). As calculated in Theory, $I_{E2} = \frac{R_1}{R_2} I_{E1} + \frac{V_{BE1} - V_{BE2}}{R_2}$.

When the difference between V_{BE1} and V_{BE2} is very small, being divided by R_2 further decreases

this difference. So, $I_{E2} = \frac{R_1}{R_2} I_{E1} = I_{E1}$ if $R_1 = R_2$.

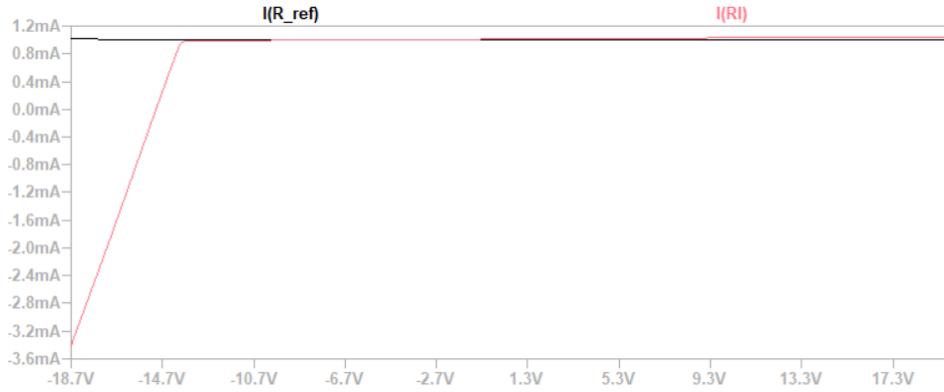


Figure 12

The simulated results are shown in **Figure 12**. I_{ref} was stable around 0.996mA. I_L increased from -3.4mA to 0.985mA linearly in the saturation region, then increased from 0.985mA to 1.047mA in the active region. Compared to Part I, the result was better as I_L was closer to I_{ref} and increased less.

Part IV. Emitter Degeneration Physical Verification

(b). In the physical circuit, I_{ref} was measured to be stable at 1.0037mA. I_L was 0.9183mA initially, then slowly increased to 1.00mA over time. The change of I_L was much smaller compared to Part II.

(d-e). With a varying V_L the measured I_L values were measured and plotted as shown in **Figure 13**. The data showed that when Q2 was in the active mode ($V_L \geq -13V$), I_L was more stable near 1mA compared to Part II. When Q2 was in the active mode, the measured results in this part were very close to the simulated results in Part III.

VL (V)	IL (mA)
-20	-3.9248
-17	-1.8676
-13	0.9697
-5	0.9938
0	1.0232
5	1.0523
13	1.135
17	1.0266
20	1.0346

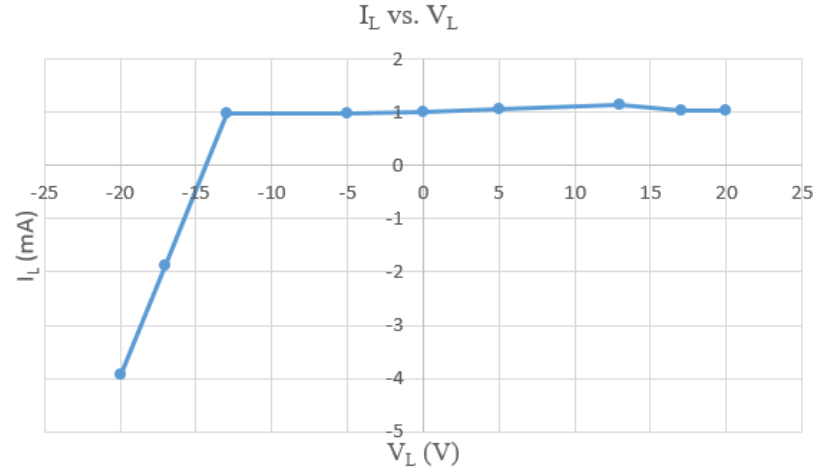


Figure 13

WEEK 2

Part I. Wilson Current Mirror Analysis and Simulation

(a). As calculated in Theory, $I_L = \frac{\beta^2 + 2\beta}{\beta^2 + 2\beta + 2} I_{ref}$.

When $\beta = 200$, $I_L = \frac{200^2 + 2 \cdot 200}{200^2 + 2 \cdot 200 + 2} I_{ref} = 0.99995 I_{ref}$. This relationship is better compared to that derived in week 1.

$$\begin{aligned}
 (b). \quad & V_L - R_L I_L - 0.7 - R_2 i_{e2} - V_{EE} > 0.3 \quad (V_{BE} = 0.7V) \\
 \Rightarrow \quad & V_L > V_{EE} + R_L I_L + 1 \quad (R_2 i_{e2} \approx 0) \\
 \Rightarrow \quad & V_L > V_{EE} + R_L I_{ref} + 1
 \end{aligned}$$

This range is narrower than the one derived in week 1.

$$(c). \quad 0 - R_{ref} I_{ref} = 0.7 + 0.7 + V_{EE} \quad \Rightarrow \quad R_{ref} = \frac{1.4 - 15}{-1m} = 13.6 \text{ k}\Omega.$$

$$(d). \quad V_{L(min)} = -15 + 1k \cdot 1m + 1 = -13 \text{ [V]}$$

(e). The simulation of the Wilson current mirror is shown in **Figure 14** below. When Q3 was in the active mode, V_L was above -13V, and both I_{ref} and I_L were stable. I_{ref} varied from 0.9992mA to 0.9997mA, while I_L varied from 0.997mA to 1.001mA. In the saturation mode, where V_L was between -15.5V and -13V, the current mirror effect disappeared and I_L was linearly related to V_L .

This was because V_C of Q3 was fixed. So, $I_L = \frac{1}{R_L} (V_L - V_C)$ is a linear relationship. As V_L dropped below 15.5V, there was no current flowing into Q1 or out of EBJ of Q3. The CBJ of Q3 functioned as a forward-biased diode. So, $I_L = \frac{1}{R_{ref} + R_L} (V_L - 0.7)$, which was a linear relationship.

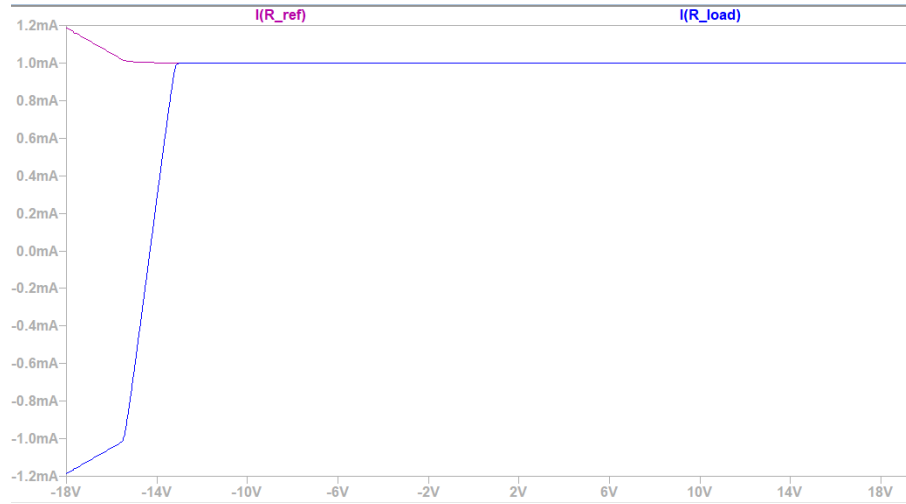


Figure 14

Part II. Current Mirror Comparison

The simulated results are shown in **Figure 15**. The starting and ending points of each circuit are shown in **Table 1**.

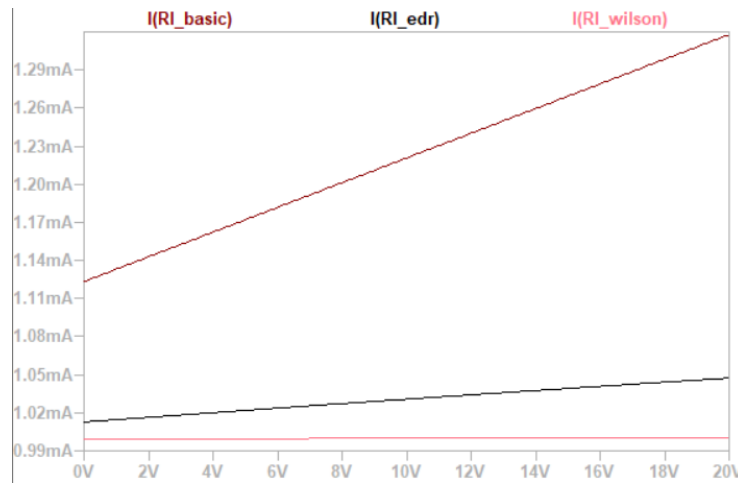


Figure 15

Table 1

	Basic	Basic with EDR	Wilson
Starting I_L (mA)	1.123	1.013	0.999
Ending I_L (mA)	1.318	1.047	1.001

The results showed that the basic circuit was the least accurate and increased the most, while the Wilson circuit was the most accurate and most stable. The Wilson circuit performed the best because all the defective factors were removed or well controlled.

Part III. Current Mirror Physical Verification

(b). In the physical circuit, I_{ref} was measured to be stable at 1.0215mA. I_L was 1.0188mA initially, then fluctuated very close to 1.0188mA over time. This result was much better than what was obtained in week 1 because I_L was more stable and closer to I_{ref} .

(d-e). With a varying V_L the measured I_L values were listed and plotted as shown in **Figure 16**.

The data showed that when Q2 was in the active mode ($V_L \geq -13V$), I_L was much more stable compared to week 1. The stability of the measured data was the same as the simulated one. However, the results were less accurate as it was not close to 1.0215mA. This could be caused by some systematic errors of the physical equipment.

V_L (V)	I_L (mA)
-20	-1.3534
-17	-1.1366
-13	0.8674
-5	0.867
0	0.8671
5	0.8659
13	0.8636
17	0.8592
20	0.8567

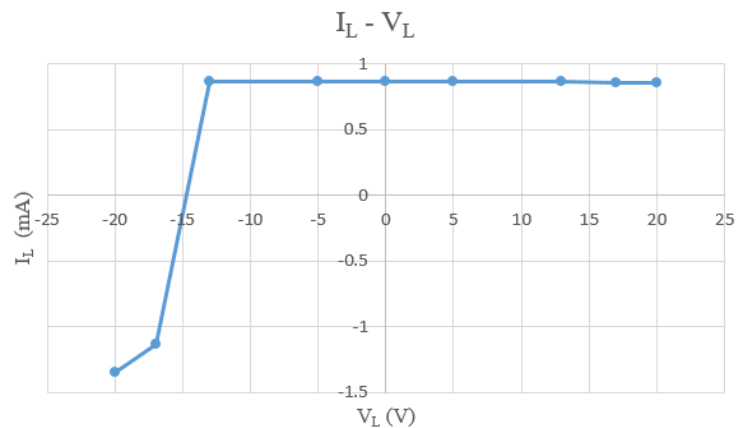


Figure 16

Conclusion

The goal of this experiment was to understand the imperfections of the basic configuration of the current mirror and implement functional current mirrors using emitter degeneration resistors and Wilson's current mirror. In the first week of the experiment, a simple NPN BJT current mirror and the one with emitter degeneration resistors were given to be simulated in LTspice and then implemented on a breadboard. The second week built upon the previous week by using concepts of BJT current mirrors to create a stable configuration using Wilson's current mirror. The circuit was simulated in LTspice and then implemented on a breadboard. Physical measurements were conducted through an oscilloscope and the Agilent Measurement Tools. Both the simulated and the measured results demonstrated that the Wilson configuration removed most of the imperfections of the other two configurations. This highlighted that the experiment was overall a success.